# Unit 2 MS-Excel and PowerPoint

## Introduction MS-Excel

Microsoft Excel is powerful spreadsheet or worksheet application which lets you create , manage, store, analyze and presents data in tabular format and also used for displaying the graphs and charts in user defined form. It is basically used to perform financial calculations, budget preparation, statistical analysis and other related operations.

Excel is an Microsoft office application that is mainly used for making calculations and mathematical works.

- Excel is an electronic spreadsheet application in which we can add sheets as per our requirements. In a single sheet, it consists of rows and columns and cells, where every cell has different address.
- Sum, product, subtraction, division & many mathematical, logical functions are available within it.
- Other features include tables, charts, clip art and more.
- It is basically used for payroll, accounts, mathematical, and for other business purposes. See below for details.

**Features/Uses of MS-Excel:-**

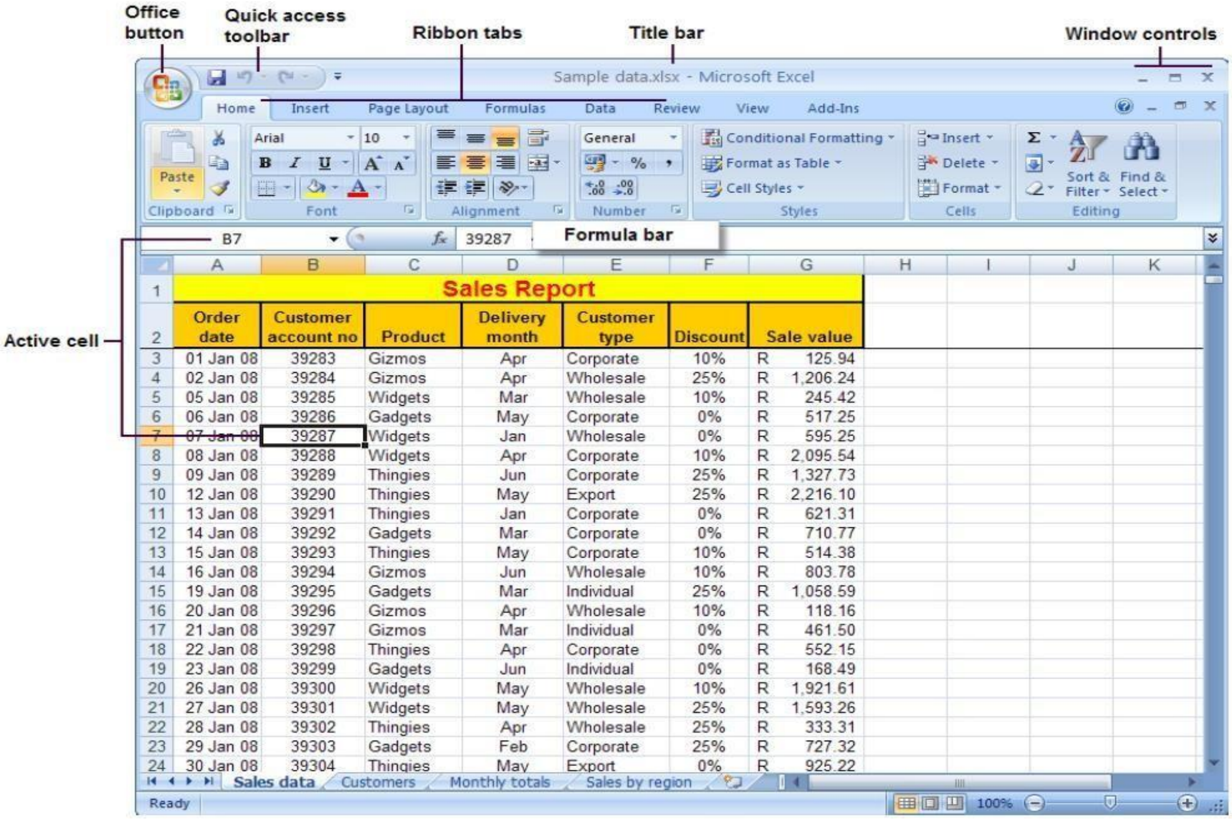
1. **Hyperlink.** We can link one file to another file or page.
2. **Clip art.** We can add images and also audio and video clips.
3. **Charts.** With charts, we can clearly show a product(s) evaluation to a client. For example, you can display a chart showing which product is selling more or less by month, week, and so forth.
4. **Tables.** Tables are created with different fields (e.g. name, age, address, roll number, and so forth). You can add a table to fill these values.
5. **Functions.** There are both mathematical functions (add, subtract, divide, multiply), and logical ones (average, sum, mod, product).
6. **Images and backgrounds.** You can incorporate images and backgrounds into each sheet.
7. **Macros.** Macros are used for recording events for future use.
8. **Database:** With the data feature, you can add any database from other sources to it.
9. **Sorting and filtering.** We can sort and/or filter our data so that anything redundant or repetitive can be removed more easily.
10. **Data validations.** This tool can help you consolidate your data.
11. **Grouping.** The grouping feature helps you both to group your data and ungroup it so that you have subtotals and so forth.
12. **Page layout.** Themes, colors, sheets, margins, size, backgrounds, breaks, print, titles, sheets height, width, scaling, grids, headings, views, bring to front of font or back alignment, and many more are available for you to lay out your page.
13. Preparation of annual reports.
14. Budgeting and forecasting
15. Accounts payable and Receivable
16. Inventory Management
17. Generating business plans
18. Tax statements.
19. Microsoft Excel allows you to enter, modify, delete and stores the data

electronically.

20. It is a very good tool for preparing wide variety of highly customizable business charts.

## Applications of Spread sheet:

- **Payment of bills**
- **Income tax calculations**
- **Invoices or bills**
- **Account Statements**
- **Inventory Control**
- **Cost-Benefits Analysis**
- **Financial Accounting**
- **Tender Evaluation**
- **Result analysis of students**




A **workbook** is a collection of **worksheets** (spreadsheets) and macros. By default, Excel **creates 3** (Sheet1, Sheet2, and Sheet3) worksheets in a new workbook.

**<u>Worksheet</u>**:-

It is also known as spreadsheet or sheet, is a rectangular grid of rows and columns.

**<u>Columns</u>**:-

In a spreadsheet column is defined as the vertical block of cells that runs through the entire worksheet. A worksheet contains 16,384 columns (In Ms- Excel 2007). The columns are identified by **letters**.

**<u>Rows</u>**:-

In a spreadsheet row is defined as the horizontal block of cells that runs through the entire width of worksheet. A worksheet contains 1,048,576 rows (In Ms-Excel 2007). The rows are identified by **numbers**.

**<u>Cell</u>**:-

The intersection of row and column creates a cell. Cells are the foundation of any spreadsheet. All data including text, numbers, and formulas are entered into spreadsheet through individual cells. Each cell is assigned a name according to its column letter and row number. Ex- D4

**Starting MS-Excel: -**
All Programs -> Microsoft Office - > Microsoft Excel

- **<u>Operation related to Workbook (File Management) : -</u>**

**Create New Workbook:-**
Office Button ->File -> New -> Blank Workbook -> Create

**To Open a Workbook:-**

1. Click the **Microsoft Office Button** tab.
2. Click **Open**.
3. Select the workbook name that you want to open.

**Saving a workbook**

6. Click the Microsoft Office button.
7. Click **Save**. The Save As dialog box appears, if you are saving your document for the first time.
8. Specify the correct folder in the Save In box.
9. Type your File Name in the File Name box.
10. Click Save.

**<u>Save As</u>** - Saves by opening a window which gives the opportunity to change the file name, location or format.

## Add a worksheet

1. Place the cursor on the tab of one of the worksheets.
2. Press the **right** mouse button.
3. From the context menu, select the **Insert** option.

## Rename a Worksheet

By default, the worksheets are named Sheet1, Sheet2 and Sheet3. To give a worksheet a more specific name, execute the following steps.

Right click on the sheet tab of Sheet1.
1. Choose Rename.
2. Type name e.g. student record

## Delete a Worksheet

To delete a worksheet, right click on a worksheet and choose Delete.

## Copy a Worksheet

1. Right click on the sheet tab of sheet1.
2. Choose Move or Copy... The 'Move or Copy' dialog box appears.
3. Select (move to end) and check Create a copy.
4. Click OK.

## Move a Worksheet

1. Right click on the sheet tab of sheet1.
2. Choose Move or Copy...
3. Select Sheet ( from before sheet)
4. Click OK

- **<u>Formatting Sheet:-</u>**

Undo
Clearing Cells
Formatting Values
Formatting Labels
Format Painter
Centering Text across Columns

- **<u>Functions and Formula:-</u>**

Excel performs calculations using formulas and functions. A formula is a series of commands instructing Excel to perform calculations based on designated values, cell references, and commands. A function is a pre-written formula. A common example of a function is *Average*, a pre-written formula which finds the sum of a set of numbers, counts how many numbers are in the set, and then divides to find the average. To save time and work, functions can be embedded within formulas.
Formulas can be used not only for small mathematical operations within cells and worksheets

*To create a simple formula that adds the contents of two cells:*

- Click the cell where the answer will appear (C5, for example).
- Type the equals sign (=) to let Excel know a formula is being defined.
- Type the cell number that contains the first number to be added (C3, for example).
- Type the **addition sign (+)** to let Excel know that an add operation is to be performed.
- Type the cell address that contains the second number to be added (C4, for example).
- Press **Enter**, or click the **Enter button** on the Formula bar to complete the formula.

**Some important mathematical functions**

1) Minimum: - Calculate the minimum from the record.

e.g. =min (a2:d2)

2) Maximum: - Calculate the maximum from the record.
e.g. =max (a2:d2)

3) Median: - Calculate the median from the record.

e.g. =median (4, 5, 6, 7, 8, 9)

4) Mode: - Calculate the mode from the record.

e.g. =mode (4, 5, 6, 7, 8, 9)

5) Sum: - Calculate the sum from the record.

e.g. = sum (a2:d2)

6) Product: - Calculate the product from the record.

e.g. =product (10, 2)

7) Power: - Calculate the power of the entered values.

e.g. =power (2, 3)

8) Mod:- Calculate the mod of the entered values.

e.g. = mod (12, 5)

9) Average: - Calculate the average of the entered values.

e.g. =average (a2:d2)

10) Fact: - Calculate the factorial of the entered values.

e.g. =Fact (4).

**String / Text Functions:-**

**CONCATENATE (text1,text2)** = Used to join 2 or more strings together

e.g. =CONCATENATE ("san","gola")

**LOWER (upper text)** Converts all letters in the specified string to lowercase

e.g =LOWER ("PUNE")

**UPPER (lowertext)** Converts all letters in the specified string to lowercase

e.g =UPPER ("pune")

**REPLACE ()** **Replaces** a sequence of characters in a string with another set of characters

e.g. =REPLACE ("solapur", 1, 2,"xy")

**SEARCH ()** Returns the **location** of a substring in a string

**e.g.** =SEARCH("a","sangli") Ans:= 2

- **<u>Charts:-</u>**

A **chart** is nothing but graphical representation of spreadsheet data. Charts can be stored on chart sheets which can be printed, viewed or edited later.

**Creating Chart's**

1) Select the information which you want in the chart, you should select Row and Column so that they will be automatically incorporated into the chart.
2) Go to **Insert** menu and click **Chart** option. Select the type of chart you want to create.

**Types of Charts**

There are different kinds of charts can be created using spreadsheet. Each one is suitable for different applications. The standard charts can be customizing and presented in different colors and shapes.

They are:

1) Column chart
2) Bar chart
3) Cylinder, Cone and Pyramid charts
4) Line chart
5) Pie chart
6) Scatter chart
7) Bubble chart
8) Radar charts
9) Area charts
10) Surface charts

**1) Column chart: -** Column chart are one of the most common types of chart. A column chart displays each data point as a vertical column, the height of which corresponds to the value. A column chart shows data changes over a period of time or illustrates comparisons among items. Typically, each data series is depicted in a different color or pattern. It is generally used for the data which is adjacent.

**2) Bar chart: -** It is similar to column charts, but the bars extends horizontally instead of vertically. A bar chart illustrates comparisons among individual items. Ti is recommended to use a column chart instead when working with times of dates.

**3) Cylinder, Cone and Pyramid charts :-** The Cone, Cylinder and Pyramid charts can be used in the same scenario as the column charts, but Use cones, cylinder or pyramids instead of rectangles. Their 3-D visual effect can enhance the overall analysis of data.

**4) Line chart: -** A line chart is used to analyze ups and downs of a tendency in a range of values. It compares trends over a period of time. Line chart is used in productions, sales or stock markets situations to show the trend of revenue or sales over a period of time.

**5) Pie chart: -** The Pie Chart is usually used to compares the size of pieces in a whole unit.

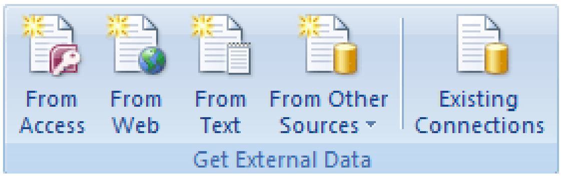
**6) Scatter chart : -** This is XY scatter chart. It is used to show correlations between two sets of values, one on the x-axis and another on the y-axis. It is good for finding patterns/trends. Scatter charts are commonly used for scientific data. This is generally not used with time, use a line chat instead.

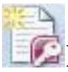
**7) Bubble chart: -** A bubble chart is a type of x y (scatter) chart. It compares set of three values and can be displayed with a 3-D visual effects. It is a variation of scatter chart in which data points are placed with bubbles, the size of bubble or indicates the value of third variable.

To arrange your data for bubble chart, place x values in one row or column and enter corresponding y values and bubble sizes in the adjacent rows or columns.

- **<u>Data Menu:-</u>**

Get External Data




From Access - Import data from a Microsoft Access database.

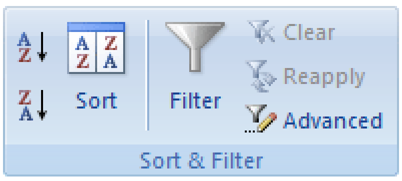
From Web - Import data from a web page.

From Text - Import data from a text file.

From Other Sources - Import data from other data sources.

Existing Connections - Connect to an external data source by selecting from a list of commonly used sources.

Sort & Filter




Sort A to Z - Sort the selection so that the

Lowest values are at the top of the column.

Sort Z to A - Sort the selection so that the highest values are at the top of the column.

Sort - Launch the Sort to sort data based on several criteria at once.

Filter - Enable filtering of the selected cells. Once filtering is turned on, click on the arrow in the column header to choose a filter for the column. The keyboard shortcut to filter is Ctrl + Shift + L.

Clear - Clear the filter and sort data for the current range of data.

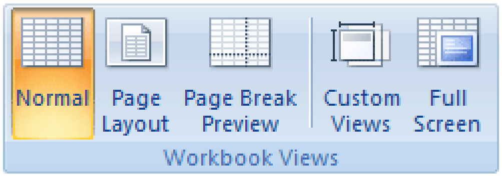
Reapply - Reapply the filter and sort in the current range. New or modified data in the column won't be filtered or sorted until you click Reapply. The keyboard shortcut to reapply is Ctrl + Alt + L.

Advanced - Specify complex criteria to limit Which records are included in the result set of a query

- **<u>Using the View Tab in Excel 2007</u>**

Tip: If you are unsure what the function of a feature is, let your cursor hover over the button (in Excel) to see a pop-up box explaining the feature. If you want to collapse the ribbon so that none of the buttons are displayed, double-click the name of the tab.

Workbook Views




Normal - View the document in Normal view.

Page Layout - View the document as it will appear on the printed page. Use this view to see where pages begin and end, and to view any headers or footers on the page.

Page Break Preview - View a preview of where pages will break when this document is printed.

Custom Views - Save a set of display and print settings as a custom view. Once

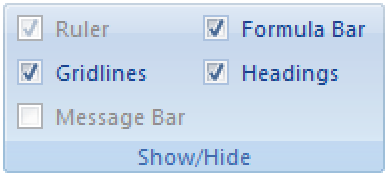
You have saved the current view, you can apply it to the document by selecting it from the list of available custom views.

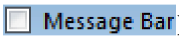
Full Screen - View the document in full screen mode.

## Show/Hide

Ruler - View the rulers used to measure and line up objects in the document.

Grid lines - Show, or hide, the lines between rows and columns in the sheet. Showing makes numbers in columns or rows easier to read or edit. Hiding Grid lines is useful if you are making a graphic organizer in Excel. These lines will not print unless the Print box is checked.



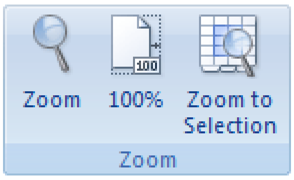
Message Bar - Open the Message Bar to complete any required actions on the document.

Formula Bar - View the formula bar in which you can enter text and formulas into cells.

Headings - Show row and column headings. Row headings are the row numbers on the side of the sheet that range from 1 to 1,048,576. Column headings are the letters that appear above the columns on a sheet that range from A to XFD. This is also found on the Page Layout tab of an Excel Workbook.

## Zoom

Zoom - Open the Zoom dialog box to specify the zoom level of the document. In most cases, you can also use the zoom controls in the status bar at the bottom right portion of the window to quickly zoom the document.




100% - Zoom the document to 100% of the normal size. Maximum Zoom 400%

Zoom to Selection - Zoom the worksheet so that the

Currently selected range of cells fills the entire window. This can help you to focus on a specific area of the worksheet.

**Error Checking**

**1.Automatic Error Checking**

**Enable Automatic Error Checking**: Go to "File" > "Options" > "Formulas" and check the box next to "Enable background error checking".

**Error Checking Rules**: Excel automatically checks for errors based on predefined rules, such as:
Formula errors (e.g., #N/A, #REF!)
Inconsistent formulas
Unprotected formulas
Calculated columns

**2.Manual Error Checking**

**Check Formulas**: Select a cell and press "F2" to edit the formula. Check for errors in the formula bar.
**Check Cell Values**: Select a cell and check its value. Ensure it matches the expected value.
**Use Error-Checking Tools**: Use tools like "Trace Precedents" and "Trace Dependents" to identify errors in formulas.

**Error Messages in Excel**

**#N/A (Not Available)**: Indicates a formula error, such as a missing value.
**#REF! (Reference)**: Indicates a formula error, such as an invalid cell reference.
**#DIV/0! (Divide by Zero)**: Indicates a formula error, such as dividing by zero.
**#VALUE! (Value)**: Indicates a formula error, such as an invalid value.

## Audit formulas in Excel:

### Formula Auditing Tools

1. **Trace Precedents**: Select a cell and go to "Formulas" > "Formula Auditing" > "Trace Precedents" to see the cells that affect the formula.
2. **Trace Dependents**: Select a cell and go to "Formulas" > "Formula Auditing" > "Trace Dependents" to see the cells that depend on the formula.
3. **Evaluate Formula**: Select a cell and go to "Formulas" > "Formula Auditing" > "Evaluate Formula" to step through the formula and see how it's evaluated.

### Formula Auditing Techniques

1. **Check Formula Syntax**: Ensure formulas are correctly formatted and syntax is correct.
2. **Check Cell References**: Ensure cell references are correct and not missing.
3. **Check Data Types**: Ensure data types are correct and compatible with formulas.
4. **Use Named Ranges**: Use named ranges to make formulas more readable and easier to maintain.
5. **Avoid Circular References**: Avoid using circular references in formulas, as they can cause errors.

## Create and use templates in Excel:

### Creating Templates

1. **Create a New Workbook**: Create a new workbook and set it up with the desired layout, formatting, and formulas.
2. **Save as Template**: Go to "File" > "Save As" and select "Excel Template" (*.xltx) as the file type.
3. **Choose a Location**: Choose a location to save the template, such as the "Templates" folder.

## Using Templates

1. **Access Templates**: Go to "File" > "New" and click on "My Templates" to access you're saved templates.
2. **Select a Template**: Select the template you want to use and click "Create".
3. **Customize the Template**: Customize the template as needed to fit your specific needs.

### Chart Customization Options

1. **Chart Title**: Add a title to the chart to describe its contents.
2. **Axis Labels**: Add labels to the x-axis and y-axis to describe the data.
3. **Legend**: Add a legend to the chart to explain the colors and symbols used.

4. **Data Labels**: Add data labels to the chart to display the values of each data point.
5. **Gridlines**: Add gridlines to the chart to help readers understand the data.

## Chart Formatting Options

1. **Chart Color**: Change the color of the chart to match your desired theme.
2. **Chart Style**: Change the style of the chart to match your desired theme.
3. **Font Size**: Change the font size of the chart title, axis labels, and legend.
4. **Font Color**: Change the font color of the chart title, axis labels, and legend.

## 6.4 Pivot Tables

To "Pivot" means something like "turning around a centre." In Excel we use pivot tables to do some sophisticated summaries and calculations on an existing table. You are able to very quickly analyse different things from different perspectives. That is probably why Microsoft has chosen to call them "pivot tables".

We will continue working with our movie file, where we will now make some quick calculations with a pivot table.

1. Make one of the cells in the table the active one. Another Tab will appear in the Ribbon called Design.
2. Click on the Design Tab in the Ribbon.
3. Click on the Summarize With Pivot Table Tab in the Ribbon.

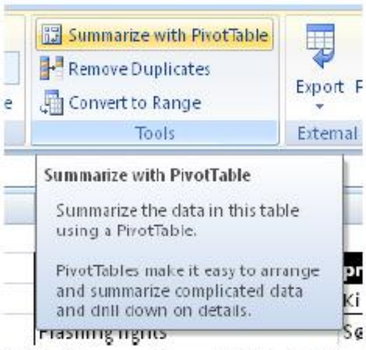



Figure 76: The Summarize with Pivot Table Tab.

4. Leave the settings as they are and click OK.

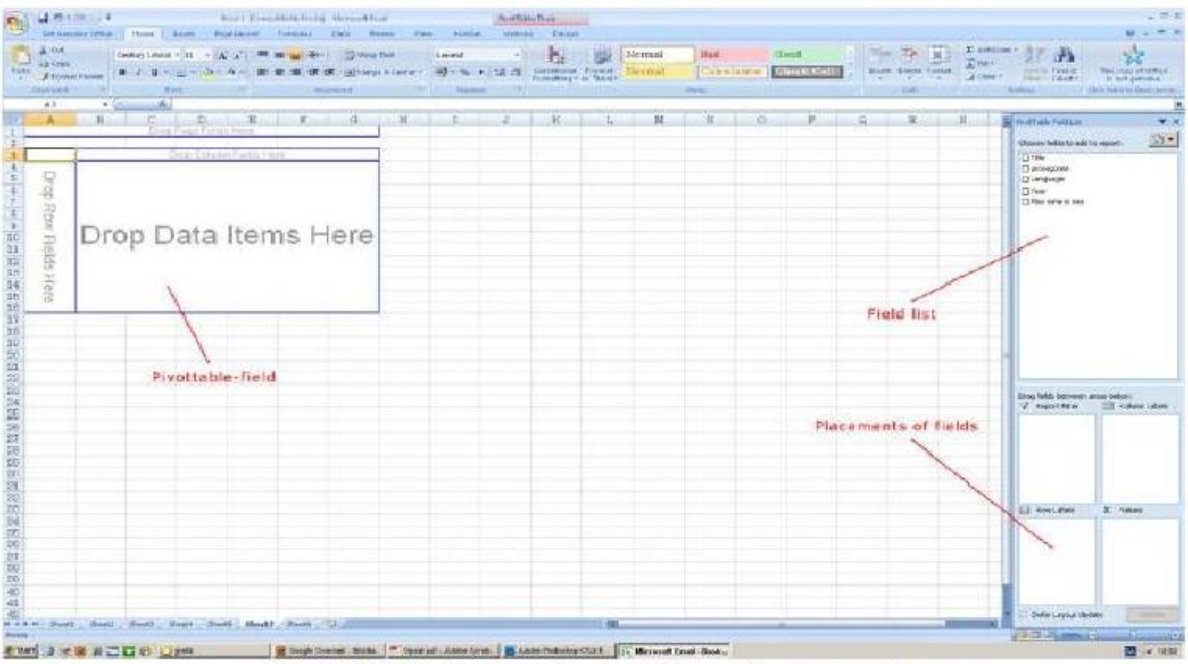



Figure 77: Work area when working with pivot tables.

- **Export data to a text file by saving it:-**

You can convert an Excel worksheet to a text file by using the **Save As** Command.

1. Click the **Microsoft Office Button**, and then click **Save As**. The **Save As** dialog box appears.
2. In the **Save as type** box, choose the text file format for the worksheet.

   For example, click **Text (Tab delimited)** or **CSV (Comma delimited).**

3. Type File Name and choose **Text (Tab delimited) and** click on **OK.**

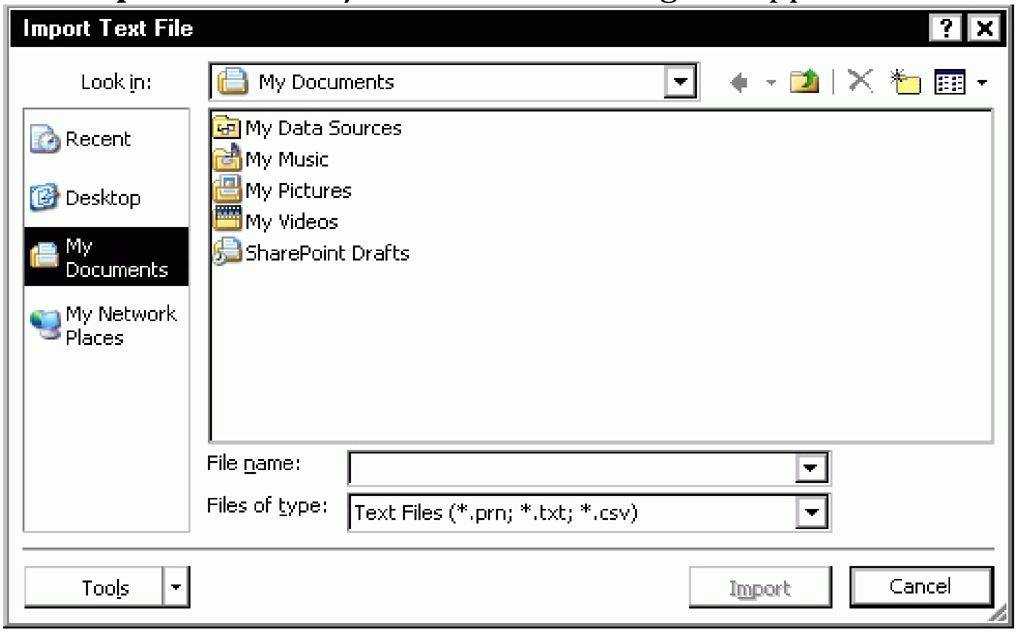
- **Importing an External Data File:-**

Importing data into an Excel worksheet is helpful if you want to use Excel to view, process and/or analyze data stored in another file.

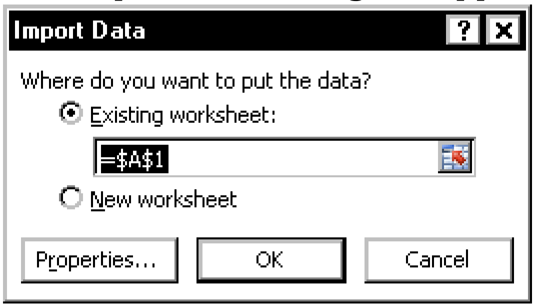
You can import data from a text file into an existing worksheet as an external data range.

1. Click the cell where you want to put the data from the text file.
2. On the **Data** tab, in the **Get External Data** group, click **From Text**.

3. The **Import Text File/Choose** a File dialog box appears.




4. The **Text Import Wizard** appears.
5. Select **Delimited** or **Fixed Width**
   NOTES:
   The Text Import Wizard automatically selects the display type that it thinks best fits your data.
   A delimiter is a character that separates pieces of data and was specified when the data was created.
6. Click **NEXT**
7. If your data is delimited, change and/or confirm the delimiters and click **NEXT**
   NOTE: The Text Import Wizard automatically selects the delimiter that it thinks is being used (usually Tab). However, you can specify a different delimiter such as, Semicolon, Comma, or Space.
8. Click **FINISH**
   The Import Data dialog box appears.




9. Click **OK**

<u>**Spreadsheet Types**</u>:

**1. Budgeting Spreadsheets**
Used for personal or business financial planning, budgeting, and expense tracking.
**2. Accounting Spreadsheets**
Utilized for financial record-keeping, invoicing, and preparing financial statements.
**3. Inventory Management Spreadsheets**
Help businesses track stock levels, monitor orders, and optimize inventory.
**4. Project Management Spreadsheets**
Used to plan, organize, and track progress of projects, including timelines and resource allocation.

**5. Data Analysis Spreadsheets** Used for Employed for data visualization, statistical analysis, and data mining.
**6. Scheduling Spreadsheets** Used for creating and managing schedules, such as employee work schedules or event timelines.
**7. Scientific Spreadsheets**
Utilized for data collection, analysis, and visualization in scientific research.
**8. Educational Spreadsheets**
Used by students and teachers for various academic purposes, such as grading, data analysis, and simulations.
**9. Business Intelligence Spreadsheets**
Employed for data analysis, reporting, and visualization to support business decision-making.
**10. Personal Finance Spreadsheets**
Used for managing personal finances, including budgeting, expense tracking, and investment tracking.
These categories often overlap, and spreadsheets can serve multiple purposes.

**Main Menu in Excel:**

**1. File Menu**
Provides options for managing files, such as:
Creating new workbooks
Opening existing workbooks
Saving workbooks
Printing workbooks
Sharing workbooks
**2. Home Menu**
Offers tools for formatting and editing cells, including:
Font and alignment options
Number formatting
Cell formatting (e.g., borders, fill)
Conditional formatting
Find and replace
**3. Insert Menu**
Allows users to add various elements to their worksheets, such as:
Tables
Illustrations (e.g., shapes, icons)
Charts and graphs
PivotTables
Pictures and online images
**4. Page Layout Menu**
Provides options for customizing the layout and design of worksheets, including:
Themes
Page setup
Print options
Gridlines and headings
**5. Formulas Menu**
Offers tools for creating and managing formulas, including:
Formula writing and editing

Function libraries (e.g., logical, text, date)
Formula auditing and error checking

**6. Data Menu**

Provides options for managing and analyzing data, including:
Data validation
Data filtering and sorting
Data grouping and outlining
Data analysis tools (e.g., pivot tables, charts)

**7. Review Menu**

Offers tools for reviewing and finalizing worksheets, including:
Spell checking, Grammar checking, Commenting and tracking changes
Protecting worksheets and workbooks.

**8. View Menu**

Provides options for customizing the Excel interface and view, including:
Worksheet views (e.g., normal, page break), Zoom and window options, Toolbar and ribbon customization
Full-screen mode.

**Using HELP in MS-Excel:**

**1. Pressing the** F1 key opens the Excel Help window, which provides access to various help topics, tutorials, and resources.

**2. Help Tab**

The Help tab is located on the ribbon in Excel. Clicking on this tab provides access to various help resources, including:
Excel Help: Opens the Excel Help window.
Contact Support: Allows users to contact Microsoft support.
Send Feedback: Enables users to provide feedback to Microsoft.
What's New: Displays information about new features and updates in Excel

**3. Excel Help Window**

The Excel Help window provides access to various help topics, tutorials, and resources. Users can browse help topics, search for specific keywords, and access additional resources.
By using these methods, users can access the help they need to learn and use MS Excel effectively.

## formatting options in Excel:

**1. Number Formatting**

Allows users to control how numbers are displayed, including:

**General, Number, Currency, Accounting, Date, Time,Percentage,Fraction**

**2. Font Formatting**

Enables users to change the appearance of text, including:
Font type, Font size, Font style (e.g., bold, italic, underline), Font color

**3. Alignment and Orientation**

Allows users to control the position and orientation of text, including:
Horizontal alignment (e.g., left, center, right), Vertical alignment (e.g., top, middle, bottom)

**4. Border and Fill Formatting**

Enables users to add borders and fill colors to cells, including:
Border style (e.g., solid, dashed, dotted), Border color, Fill color, Pattern

**5. Protection Formatting**

Allows users to protect cells and worksheets from editing, including:
Locking cells,Hiding formulas,Protecting worksheets

**6. Conditional Formatting**

Enables users to apply formatting based on conditions, including:
Highlighting cells based on values,Creating data bars and color scales Applying icons and graphics.

**7. Shortcut Keys for Formatting**

Provides users with shortcut keys for common formatting tasks, including:
Ctrl + B (bold)
Ctrl + I (italic)
Ctrl + U (underline)
Ctrl + 1 (apply general number format)

**8. Format Painter**
Allows users to copy formatting from one cell or range to another.
**9. Clear Formatting**
Enables users to remove all formatting from a cell or range.
**10. Format Cells Dialog Box**
Provides users with a comprehensive dialog box for formatting cells, including options for number, font, alignment, border, fill, and protection.

## Format a sheet in Excel:

1**. Rename a Sheet:** Right-click on the sheet tab and select "Rename" to give the sheet a new name.
**2. Insert a New Sheet:**
Click on the "+" icon next to the existing sheet tabs to insert a new sheet.
**3. Delete a Sheet**
Right-click on the sheet tab and select "Delete" to remove the sheet.
**4. Change Sheet Tab Color**
Right-click on the sheet tab and select "Tab Color" to change the color of the tab.
**5. Protect a Sheet**
Go to the "Review" tab and click on "Protect Sheet" to protect the sheet from editing.
**6. Hide a Sheet**
Right-click on the sheet tab and select "Hide" to hide the sheet.
**7. Unhide a Sheet**
Go to the "Home" tab, click on "Format", and select "Hide & Unhide" to unhide a hidden sheet.
**8. Change Sheet Orientation**
Go to the "Page Layout" tab and click on "Orientation" to change the sheet orientation.
**9. Set Print Area**
Go to the "Page Layout" tab and click on "Print Area" to set the print area for the sheet.
**10. Freeze Panes**
Go to the "View" tab and click on "Freeze Panes" to freeze rows or columns in the sheet.
By using these formatting options, you can customize and organize your Excel sheets to better suit your needs.

## Format a row in Excel:

**1. Row Height**
Select the row and drag the row border to adjust the row height.
**2. Row Width**
Select the row and drag the column border to adjust the row width.
**3. Font Formatting**
Select the row and use the font tools in the "Home" tab to change:
Font type, Font size, Font style (e.g., bold, italic, underline),Font color
**4. Alignment and Orientation**
Select the row and use the alignment tools in the "Home" tab to change:
Horizontal alignment (e.g., left, center, right)
Vertical alignment (e.g., top, middle, bottom)
**5. Border and Fill Formatting**
Select the row and use the border and fill tools in the "Home" tab to change:
Border style (e.g., solid, dashed, dotted)
Border color, Fill color, Pattern.
**6. Row Color**
Select the row and use the "Fill Color" tool in the "Home" tab to change the row color.
**7. Hide and Unhide Rows**
Select the row and right-click to hide or unhide the row.
**8. Row Formatting Using Shortcuts**
Ctrl + Shift + F (format as table)
Ctrl + Shift + > (increase font size)
Ctrl + Shift + < (decrease font size)
By using these formatting options, you can customize the appearance of your rows in Excel.

## FORMATTING COLUMN

### 1. Column Width

Select the column and drag the column border to adjust the column width.

### 2. AutoFit Column Width

Double-click on the column border to AutoFit the column width.

### 3. Column Formatting

Select the column and use the formatting tools in the "Home" tab to change:
Font type, Font size, Font style (e.g., bold, italic, underline), Font color

### 4. Alignment and Orientation

Select the column and use the alignment tools in the "Home" tab to change:
Horizontal alignment (e.g., left, center, right)
Vertical alignment (e.g., top, middle, bottom)
Orientation (e.g., angle, rotate)

### 5. Border and Fill Formatting

Select the column and use the border and fill tools in the "Home" tab to change:
Border style (e.g., solid, dashed, dotted), Border color, Fill color, Pattern

### 6. Column Color

Select the column and use the "Fill Color" tool in the "Home" tab to change the column color.

### 7. Hide and Unhide Columns

Select the column and right-click to hide or unhide the column.

### 8. Column Formatting Using Shortcuts

Ctrl + Shift + F (format as table)
Ctrl + Shift + > (increase font size)
Ctrl + Shift + < (decrease font size)
By using these formatting options, you can customize the appearance of your columns in Excel.

## Hiding AND Locking Cells

**Hide Formulas**: Select the cells containing formulas, go to the "Home" tab, click on "Format" in the "Cells" group, and select "Hide & Unhide" > "Hide Formulas".
**Hide Rows or Columns**: Select the rows or columns you want to hide, right-click, and select "Hide".
**Hide Sheets**: Right-click on the sheet tab and select "Hide".

## Locking Cells:

**Lock Cells**: Select the cells you want to lock, go to the "Home" tab, click on "Format" in the "Cells" group, and select "Lock Cells".
**Protect Worksheet**: Go to the "Review" tab, click on "Protect Worksheet", and select the options you want to allow or disallow.
**Password-Protect Worksheet**: Go to the "Review" tab, click on "Protect Worksheet", and select "Encrypt with Password".

## Row Headers AND Column Headers

**Row Headers**: Row headers are displayed by default in Excel. If they are hidden, go to the "View" tab and select "Headings" to display them.
**Hiding Row Headers**: To hide row headers, go to the "View" tab and deselect "Headings".

### Column Headers:

**Displaying Column Headers**: Column headers are displayed by default in Excel. If they are hidden, go to the "View" tab and select "Headings" to display them.
**Hiding Column Headers**: To hide column headers, go to the "View" tab and deselect "Headings".

## Sheet Formatting:

**Sheet Background**: Right-click on the sheet tab, select "View Code", and add a background image or color.
**Sheet Color**: Right-click on the sheet tab and select "Tab Color" to change the sheet tab color.
**Sheet Protection**: Go to the "Review" tab and click on "Protect Sheet" to protect the sheet from editing.

## Cell Styles:

**Predefined Cell Styles**: Go to the "Home" tab and click on "Cell Styles" to apply predefined cell styles.
**Custom Cell Styles**: Go to the "Home" tab, click on "Cell Styles", and select "New Cell Style" to create a custom cell style.

### Adding a Background Image

1. **Select the Sheet**: Click on the sheet tab to select the sheet.
2. **Go to Page Layout**: Click on the "Page Layout" tab in the ribbon.
3. **Click on Background**: Click on the "Background" button in the "Page Setup" group.
4. **Select a Picture**: Select a picture from your computer or online.
5. **Insert the Picture**: Click "Insert" to add the picture as the background.

### Changing the Sheet Tab Color

1. **Right-Click on the Sheet Tab**: Right-click on the sheet tab to open the context menu.
2. **Select Tab Color**: Select "Tab Color" from the context menu.
3. **Choose a Color**: Choose a color from the palette or enter a custom color code.

### Changing the Background Color of Cells

1. **Select the Cells**: Select the cells you want to change the background color for.
2. **Go to Home**: Click on the "Home" tab in the ribbon.
3. **Click on Fill Color**: Click on the "Fill Color" button in the "Font" group.
4. **Choose a Color**: Choose a color from the palette or enter a custom color code.

### Changing the Font Color of Cells

1. **Select the Cells**: Select the cells you want to change the font color for.
2. **Go to Home**: Click on the "Home" tab in the ribbon.
3. **Click on Font Color**: Click on the "Font Color" button in the "Font" group.
4. **Choose a Color**: Choose a color from the palette or enter a custom color code.

### Here's how to work with borders and shading in Excel:

### Borders

1. **Select Cells**: Select the cells you want to add borders to.
2. **Go to Home**: Click on the "Home" tab in the ribbon.
3. **Click on Borders**: Click on the "Borders" button in the "Font" group.
4. **Choose a Border Style**: Choose a border style from the drop-down menu.
5. **Customize Borders**: Use the "Borders" tab in the "Format Cells" dialog box to customize border styles, colors, and widths.

### Shading

1. **Select Cells**: Select the cells you want to add shading to.
2. **Go to Home**: Click on the "Home" tab in the ribbon.
3. **Click on Fill Color**: Click on the "Fill Color" button in the "Font" group.
4. **Choose a Fill Color**: Choose a fill color from the palette or enter a custom color code.
5. **Customize Shading**: Use the "Fill" tab in the "Format Cells" dialog box to customize fill colors, patterns, and gradients.

### Gradient Fill

1. **Select Cells**: Select the cells you want to add a gradient fill to.
2. **Go to Home**: Click on the "Home" tab in the ribbon.
3. **Click on Fill Color**: Click on the "Fill Color" button in the "Font" group.
4. **Choose a Gradient Fill**: Choose a gradient fill from the "Gradient" tab in the "Format Cells" dialog box.
5. **Customize Gradient Fill**: Customize the gradient fill by adjusting the colors, direction, and style.

   **Anchoring Objects to Cells:** Anchoring the object will be either move, resize, or stay in a fixed position relative to the cells.

Steps: **Insert an Object** (e.g., a picture, shape, or chart):

1. Go to the **Insert** tab on the ribbon.
2. Choose the type of object (e.g., Picture, Shape, Chart, Text Box, etc.) and insert it.
3. **Select the Object** you want to anchor.
4. **Right-click on the Object** and choose **Format Object** (the option might vary based on the object).
5. In the dialog box, go to the **Properties** tab.
6. Select the option **"Move and size with cells"**:
7. **Move and size with cells**: This will anchor the object to the cells. If you adjust the size or move the cells, the object will follow.
8. **Move but don’t size with cells**: The object moves when cells are moved, but it doesn't resize.
9. **Don’t move or size with cells**: The object stays in the same place, no matter what happens to the cells.

10. Click **OK** to apply the settings.

**Track Changes in Excel:**

This feature allows you to highlight changes made by different users in a shared workbook.

**Steps to Track Changes:**

1. Open your workbook in Excel.
2. Go to the Review tab.
3. Click on Track Changes (in older versions of Excel) or choose Share Workbook.
4. In the dialog box, select Track changes while editing and then choose how you want to display changes.
5. Click OK to confirm your settings.
6. Afterward, changes made will be highlighted with different colours, and you can view who made each change.

**Security in MS Excel:**

Microsoft Excel provides a variety of **security features** to protect sensitive data and ensure that workbooks are only accessible to authorized users.

**How to Set Password Protection:**

1. Open your workbook and go to File > Info.
2. Click Protect Workbook.
3. Select Encrypt with Password (to protect from unauthorized access).

4 Enter your desired password, then confirm it

## Subtotal in Excel

The **Subtotal** feature in Excel is used to add subtotals to data that is grouped by a specific column, allowing for easier analysis of different segments of data. This feature works well with sorted data.

Use Subtotal Feature:

1. Go to the Data tab, in the Outline group, and click Subtotal.
2. In the Subtotal dialog box:
3. In the At each change in drop-down, select the column you want to group by (e.g., "Department").
4. In the Use function drop-down, choose the function you want (e.g., Sum, Average, Count, etc.).
5. In the Add subtotal to section, select the columns you want to subtotal (e.g., Sales).
6. Click OK.

**Data Validation in Excel**

Data Validation is a feature in Excel that allows you to control the type of data entered into a cell, preventing errors and ensuring consistency. You can set rules for what type of data (numbers, text, dates, etc.) is allowed in a given range of cells.

**Steps of Data Validation:**

1. **Select the cell or range** where you want to apply data validation.
2. Go to the **Data** tab in the ribbon.
3. Click **Data Validation** (in the **Data Tools** group).
4. In the **Data Validation** dialog box, under the **Settings** tab, select the type of validation you want:
5. **Whole Number**: Restrict the input to whole numbers within a specified range (e.g., 1 to 100).
6. **Decimal**: Restrict the input to decimal values within a range.